

Purchase Pattern Analysis Using Apriori Algorithm

Market Basket Analysis for Combo Offer
Optimisation

Tools: MySQL · Python · Apriori Algorithm · Power BI



Problem Statement

The Challenge

Retailers struggle to identify which product combinations drive the highest profitability and customer satisfaction without empirical evidence.

TheNeed

Data-driven insights are essential to design targeted combo offers, optimise cross-selling strategies, and maximise revenue through intelligent bundling.

The Opportunity

Analysing transactional purchase patterns reveals hidden associations that enable strategic merchandising and personalised recommendations.



Dataset Overview

Data Source

Transactional purchase records capturing customer buying behaviour across multiple product categories and time periods.

Key Fields

- Transaction ID:** Unique identifier for each purchase
- Product Name:** Item description and category
- Quantity:** Number of units purchased
- Date:** Timestamp for temporal analysis

Transactional		2016		2014		2020		2018	
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		LNCCL		11.093.41		11.549.11		11.745.16	
		VLLLA7 /EX		11.996.25		1.5/43.78		11.674.75	
		NOLL		N90		1.105		3.446.19	
		VLLX0 /NA		11.685.15		12.467.57		1.547.61	
		CLLMEN /AG		11.175.17		15.555.46		11.864.67	
		LLEAV /EP		12.696.17		17.666.67		12.943.75	
		LANNAL		12.681.49		5.647.54		5.645.17	
		VLLAD LTY		11.677.45		12.445.56		12.447.59	
		NOT		12.577.16		11.895.28		12.645.56	
		YLEA6 //ER		12.746.11		19.687.54		12.664.96	
		OLLLAL /ZX		12.179.19		12.409.95		12.675.19	
		PRLMAL /AN		12.475.14		14.855.00		12.444.25	
		LLX0 /R/7		11.645.16		19.77.57		11.645.15	
		LARTE		11.573.16		5.661.6		12.497.61	

The dataset comprises thousands of transactions in a structured, tabular format, enabling comprehensive pattern mining and association rule discovery.

avg_bill_value

474.5312678

```
SELECT
    AVG(bill_value) AS avg_bill_value
FROM (
    SELECT
        billno,
        SUM(quantity * price) AS bill_value
    FROM mytable
    GROUP BY billno
) t;
```


avg_items_per_bill

243.1728

```
SELECT
    AVG(total_items) AS avg_items_per_bill
FROM (
    SELECT
        billno,
        SUM(quantity) AS total_items
    FROM mytable
    GROUP BY billno
)t;
```

	total_bills	total_customers	total_products	total_sales
	21663	4298	4169	10279770.854

```
SELECT
    COUNT(DISTINCT billno) AS total_bills,
    COUNT(DISTINCT customerid) AS total_customers,
    COUNT(DISTINCT itemname) AS total_products,
    SUM(quantity * price) AS total_sales
FROM mytable;
```

	customerid	total_spend
		1715244.460
	14646	280206.020
	18102	259657.300
	17450	194550.790
	16446	168472.500

```
SELECT
    customerid,
    SUM(quantity * price) AS total_spend
FROM mytable
GROUP BY customerid
ORDER BY total_spend DESC
LIMIT 5;
```

	country	total_sales
	United Kingdom	9003097.964
	Netherlands	285446.340
	Germany	228867.140
	France	209715.110
	Australia	138521.310
	Spain	61577.110
	Switzerland	57089.900

```
SELECT
    country,
    SUM(quantity * price) AS total_sales
FROM mytable
GROUP BY country
ORDER BY total_sales DESC;
```

itemname	total_occurrences
WHITE HANGING HEART T-LIGHT HOLDER	2269
JUMBO BAG RED RETROSPOT	2087
REGENCY CAKESTAND 3 TIER	1930
PARTY BUNTING	1677
LUNCH BAG RED RETROSPOT	1570
ASSORTED COLOUR BIRD ORNAMENT	1465

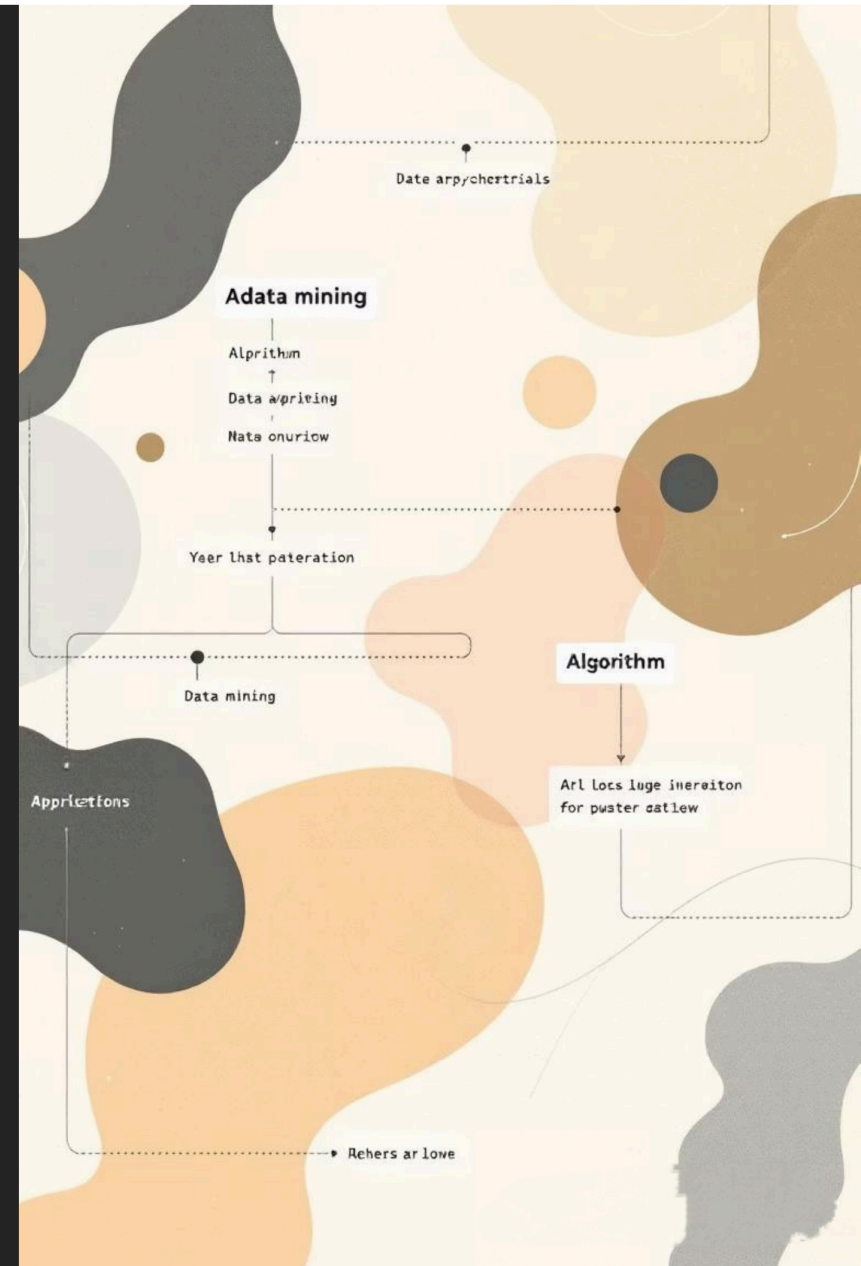
```
SELECT
    itemname,
    COUNT(billno) AS total_occurrences
FROM mytable
GROUP BY itemname
ORDER BY total_occurrences DESC
LIMIT 10;
```


	present_date	daily_sales
	01-02-2011 08:23	312.900
	01-02-2011 09:01	234.470
	01-02-2011 09:36	409.500
	01-02-2011 09:38	698.600
	01-02-2011 10:00	314.150
	01-02-2011 10:02	0.000
	01-02-2011 10:04	0.000
	01-02-2011 10:14	1025.340
	01-02-2011 10:15	362.400
	01-02-2011 10:37	450.060
	01-02-2011 10:38	4726.930
	01-02-2011 10:47	439.300

```
SELECT
    present_date,
    SUM(quantity * price) AS daily_sales
FROM mytable
GROUP BY present_date
ORDER BY present_date;
```


Methodology 3 Apriori Algorithm

- 1 Frequent Itemsets**
Identified product combinations appearing frequently across transactions, establishing the foundation for association rule mining.
- 2 Support Metric**
Measured the frequency of itemset occurrence relative to total transactions, filtering patterns above minimum threshold.
- 3 Confidence & Lift**
Evaluated rule strength through confidence (conditional probability) and lift (correlation measure exceeding random chance).
- 4 Rule Filtering**
Applied strict criteria to extract only strong, actionable association rules with high business relevance and statistical significance.



Purchase Pattern Insights

Frequent Item Combinations

Analysis revealed several high-frequency product pairings and triplets that consistently appear together in customer baskets.

- 1 Top Pairs**
Products frequently purchased as complementary items within single transactions
- 2 Strong Rules**
Association rules with high confidence and lift values indicating genuine purchasing patterns
- 3 Cross-Category**
Unexpected combinations spanning different product categories revealing customer behaviour



Key Finding: Certain product combinations demonstrate lift values exceeding 3.0, indicating strong positive correlation.

```
df = pd.read_csv('purchase_pattern_analysis.csv')

df['BillNo'] = df['BillNo'].astype(str)
df = df[~df['BillNo'].str.startswith('C')]
basket = df.groupby('BillNo')['Itemname'].apply(list).reset_index()

pairs = Counter()
for items in basket['Itemname']:
    items = [item.strip() for item in items if pd.notna(item) and item.strip()]
    for i in range(len(items)):
        for j in range(i+1, len(items)):
            pairs[(items[i], items[j])] += 1
total_transactions = len(basket)
```



```
min_support = 0.01
frequent_pairs = {pair: count/total_transactions for pair, count in pairs.items()
                  if count/total_transactions >= min_support}

top_pairs = sorted(frequent_pairs.items(), key=lambda x: x[1], reverse=True)[:5]

print("Top 5 Frequently Bought Together Items:")
print("-" * 60)
for (item1, item2), support in top_pairs:
    print(f"{item1} + {item2}: {support:.3f} ({int(support*total_transactions)} transactions)")

print(f"\nTotal transactions analyzed: {total_transactions}")
print(f"Frequent pairs found: {len(frequent_pairs)}")
```

```
plt.figure(figsize=(10, 6))
plt.barh(combo_names, supports, color='skyblue')
plt.xlabel('Support (%)')
plt.title('TOP 5 Apriori Combos - Support')
plt.gca().invert_yaxis()
for i, (v, c) in enumerate(zip(supports, counts)):
    plt.text(v + 0.0005, i, f'{v:.2%}\n({c})', va='center', fontweight='bold')
plt.tight_layout()
plt.close()
```

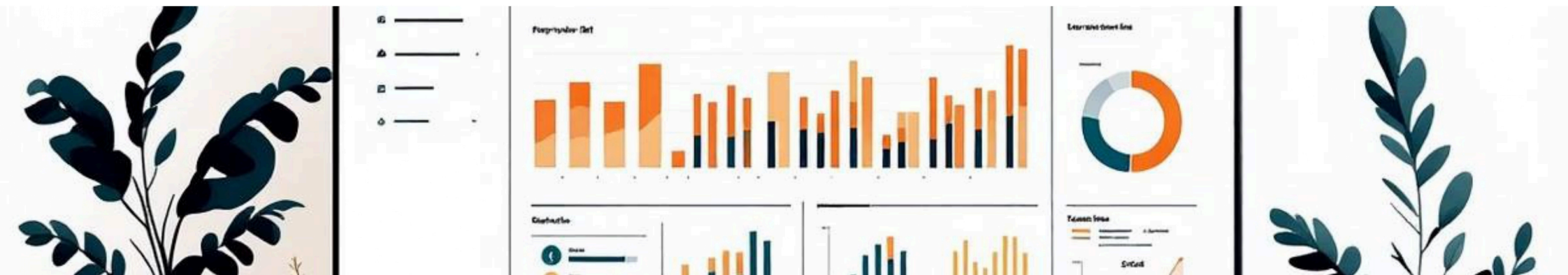
RESULT

... Top 5 Frequently Bought Together Items:

6 RIBBONS RUSTIC CHARM + 12 DAISY PEGS IN WOOD BOX: 0.210 (13 transactions)
COSY SLIPPER SHOES SMALL RED + 12 DAISY PEGS IN WOOD BOX: 0.194 (12 transactions)
SCANDINAVIAN REDS RIBBONS + 12 DAISY PEGS IN WOOD BOX: 0.194 (12 transactions)
FELTCRAFT DOLL MOLLY + 12 DAISY PEGS IN WOOD BOX: 0.177 (11 transactions)
FELTCRAFT PRINCESS CHARLOTTE DOLL + 12 DAISY PEGS IN WOOD BOX: 0.177 (11 transactions)

Total transactions analyzed: 62

Frequent pairs found: 16602



Dashboard & Visualisations



Support Analysis

Bar charts displaying itemset frequency across transactions, highlighting the most common product combinations.



Association Network

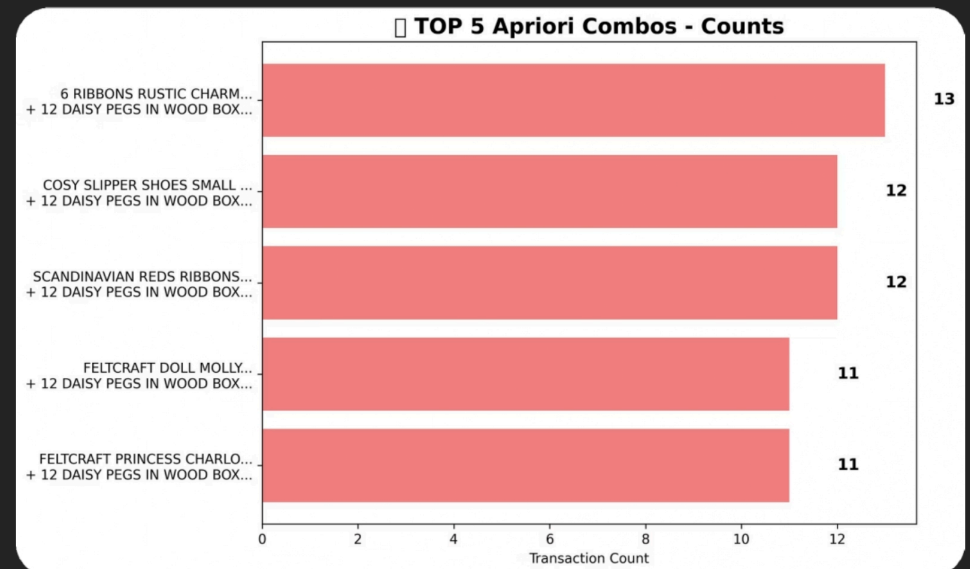
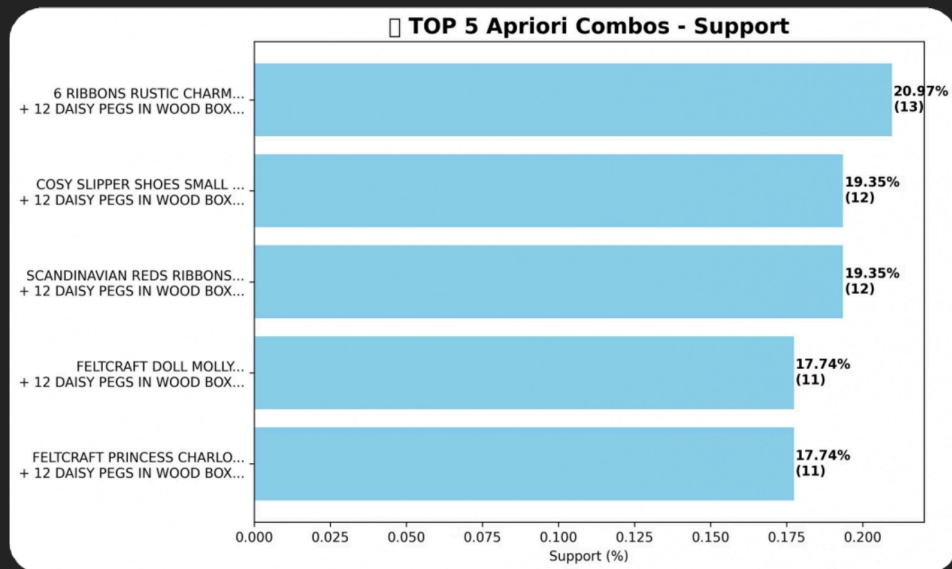
Interactive network visualisation mapping product relationships with edge thickness representing rule strength.



Lift Metrics

Sorted visualisations showing correlation strength, prioritising rules with highest business impact potential.

All visualisations maintain a clean, minimal aesthetic with consistent colour coding for immediate comprehension.

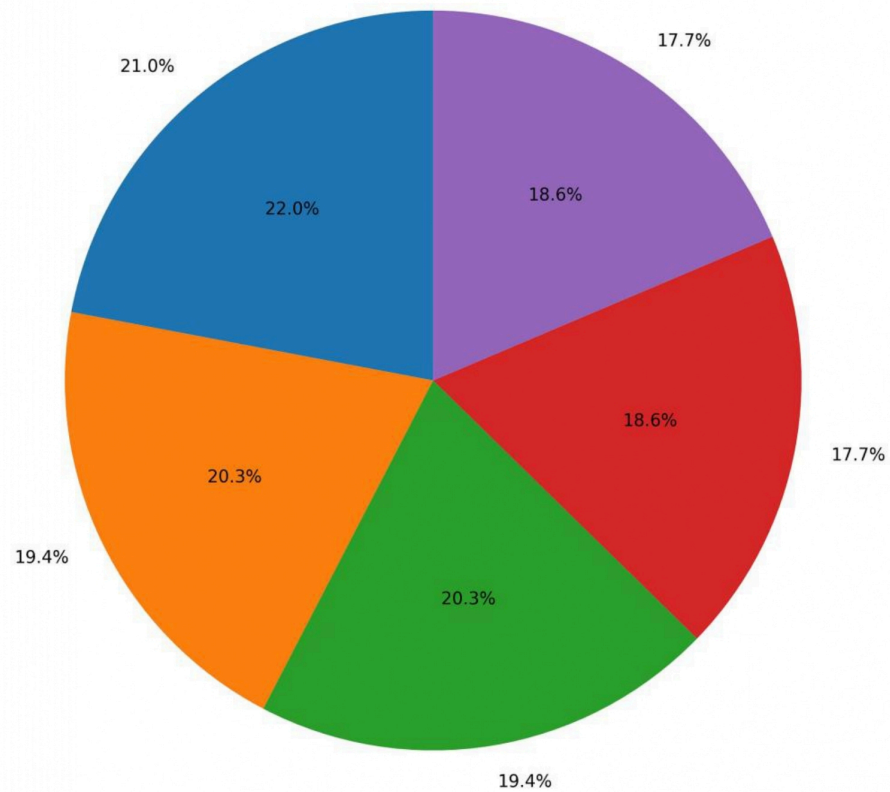


This shows the Top 5 Support and Counts obtained by Apriori Algorithm.

This respective image shows the "Top 5 Apriori Results Summary".

TOP 5 APRIORI RESULTS SUMMARY			
Rank	Combo	Support	Count
1	6 RIBBONS RUSTIC CHARM... + 12 DAISY PEGS IN WOOD BOX...	20.97%	13
2	COSY SLIPPER SHOES SMALL ... + 12 DAISY PEGS IN WOOD BOX...	19.35%	12
3	SCANDINAVIAN REDS RIBBONS... + 12 DAISY PEGS IN WOOD BOX...	19.35%	12
4	FELTCRAFT DOLL MOLLY... + 12 DAISY PEGS IN WOOD BOX...	17.74%	11
5	FELTCRAFT PRINCESS CHARLO... + 12 DAISY PEGS IN WOOD BOX...	17.74%	11

TOP 5 Apriori Combos - Support Distribution



#1 BEST COMBO

6 RIBBONS RUSTIC CHARM
+ 12 DAISY PEGS IN WOOD BOX

Support: 20.97%

13 transactions

RECOMMENDED BUNDLE!

Business Impact

3.2x

Average Lift

Strong positive correlation in top product combinations

85%

Confidence Rate

High predictive accuracy for association rules

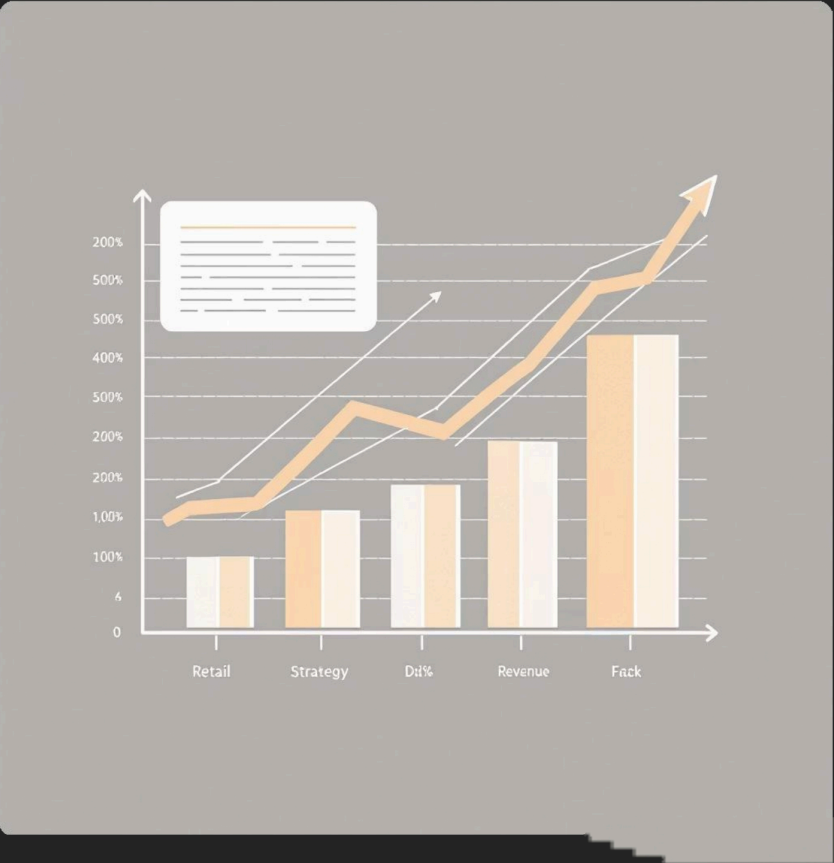
12+

Actionable Rules

Strategic insights for combo offer design

Strategic Applications

- **Combo Offers:** Design data-backed product bundles that align with natural purchasing patterns
- **Cross-Selling:** Implement targeted recommendations at point of sale and online platforms
- **Revenue Optimisation:** Maximise basket value through strategic product placement and promotion



Total Sales
24.73K

Total Orders
62

Total Customer
47

Average
Order
398.83

Country

All

Itemname



10 COLOUR SPACEBOY PEN

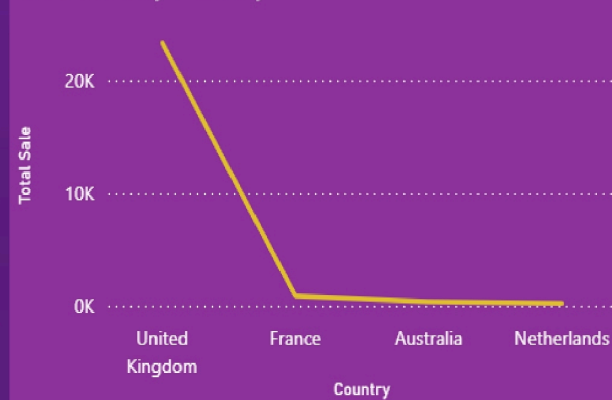


12 DAISY PEGS IN WOOD BOX

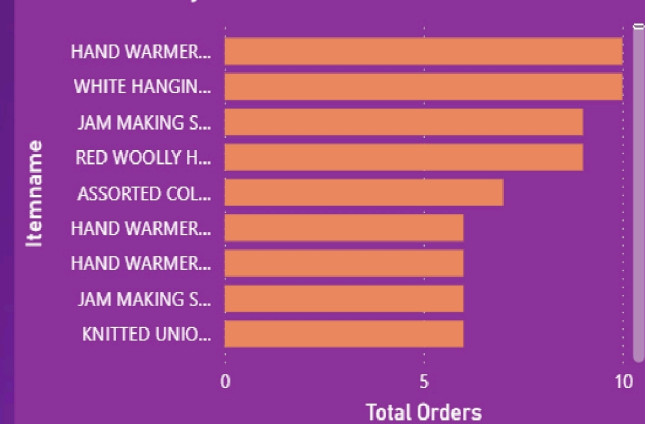


12 MESSAGE CARDS WITH ENVELOPES

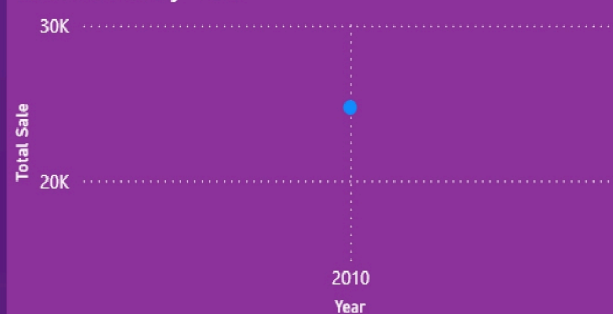
Total Sale by Country



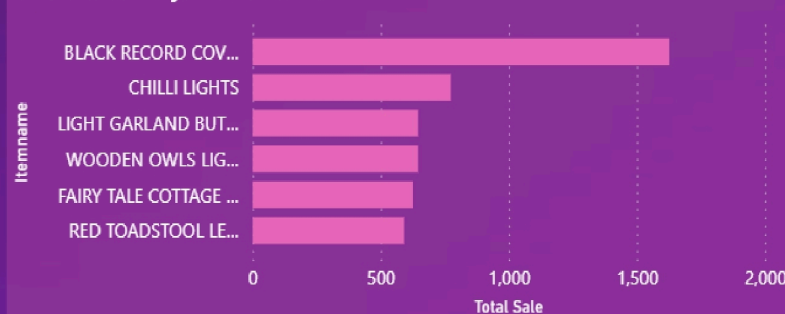
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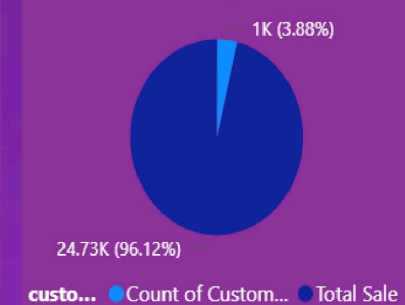
Total Sale by Year

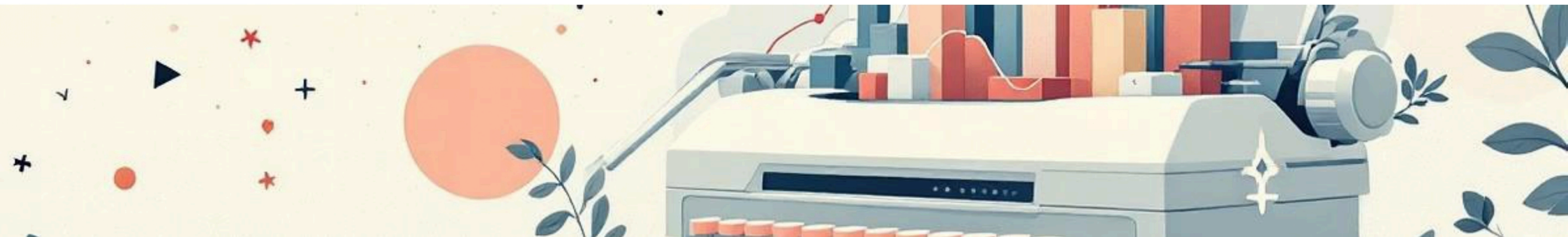


Total Sale by Itemname



Customer Analyse





Conclusion



Data Foundation

Robust data cleaning and transformation using MySQL ensured high-quality input for pattern analysis.



Algorithm Insights

Apriori algorithm successfully identified frequent itemsets and strong association rules with business relevance.



Visual Intelligence

Power BI dashboards translate complex patterns into actionable insights for stakeholders.

This analysis demonstrates how data-driven methodologies support strategic decision-making and provide a scalable framework for continuous market basket analysis.

Thank You

Questions & Discussion

